INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

(School of University of Engineering and Management, Kolkata)

End Semester Examination, November - December, 2025

Degree: **B.C.A** Semester: **First**

Stream: **BCA**

Paper Code: **BCACC102**

Paper Name: **Digital Electronics**

Full Marks: **100** Duration: **3 Hours**

Part - A

Attempt any 10 questions out of 15 questions

Each question carries 2 Marks (2 × 10)

1. Convert the binary number $(101111.1101)_2$ into its decimal equivalent. **2**

2. Convert the octal number $(120)_8$ in decimal. **2**

3. Simplify the given expression $A + A.\bar{B} + \bar{A}.B$. **2**

4. Convert $(A3B)_{16}$ to a decimal number. **2**

5. Divide the binary number 11101 by 1100. **2**

6. Express the decimal number 1993 in 5421 codes. **2**

7. Explain how the basic gates can be realised using NOR gates. **2**

8. Realise the logic expression $Y = A \oplus B \oplus C$ using the basic gates. **2**

9. Using only NAND gates, design a circuit to provide an output of logic 1 when only one of three inputs is logic 1. **2**

10. What are some of the major applications of multiplexers? **2**

11. ABCD seven segment decoder/driver is connected to an LED display. Which segment is illuminated for DCBA=0111? **2**

12. Using the conversion logic of Binary to Gray convert 1000001001 to Gray code. **2**

13. Show the logic required to convert a 10-bit Gray code to binary. 2

14. What is a modulus counter? 2

15. Explain the difference between the performance of asynchronous and synchronous counters. 2

Part - B

Attempt any 6 questions out of 9 questions

Each question carries 5 Marks (5 × 6)

16. Simplify the expression $Y = \overline{(AB + \bar{C})}\overline{(A + \bar{B} + C)}$. Using Boolean algebra. 5

17. Prove the following Boolean expression 5
$(A + B)(\bar{A}\bar{C} + C)(\overline{\bar{B} + AC}) = \bar{A}B$

18. Simplify the expression $Y = m_1 + m_5 + m_{10} + m_{11} + m_{12} + m_{13} + m_{15}$, using the *K*- map method. 5

19. Add the following numbers in 2's complement method. 5
(a) -48 and +31 (b) +38 and -22

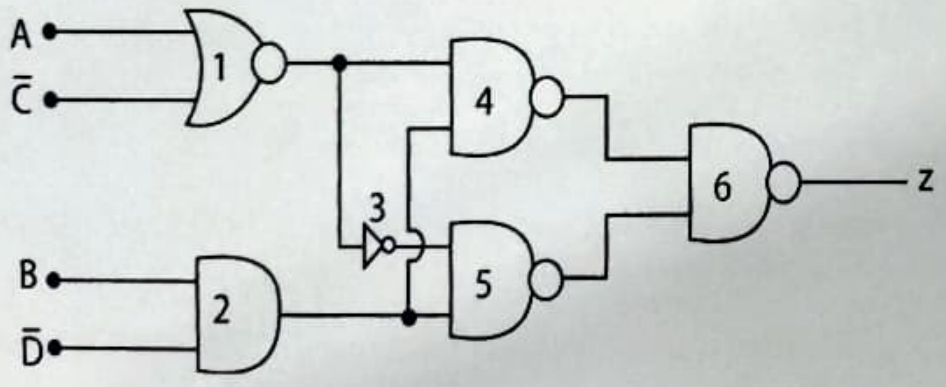
20. Design a full-subtractor using only NAND gates. 5

21. What is the major difference in the operation of edge-triggered flip-flops and master slave flip-flops? 5

22. Write a short note on Ring counter. 5

23. Differentiate between Ripple counter and synchronous counter. 5

24. Simplify the following logic circuit. 5



Part - C

Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. Implement the following Boolean function with a 4-to-1 MUX and external gates. Connect inputs A and B to the selection lines. The input requirements for the four data lines will be a function of variables C and D. These values are obtained by expressing F as a function of C and D for each of the four cases when AB – 00, 01, 10 and 11. These functions may have to be implemented with external gates. **10**

$$F(A,B,C,D) = \Sigma(1,3,4,11,12,13,15)$$

26. Design a sequence generator to generate the sequence 1001. **10**

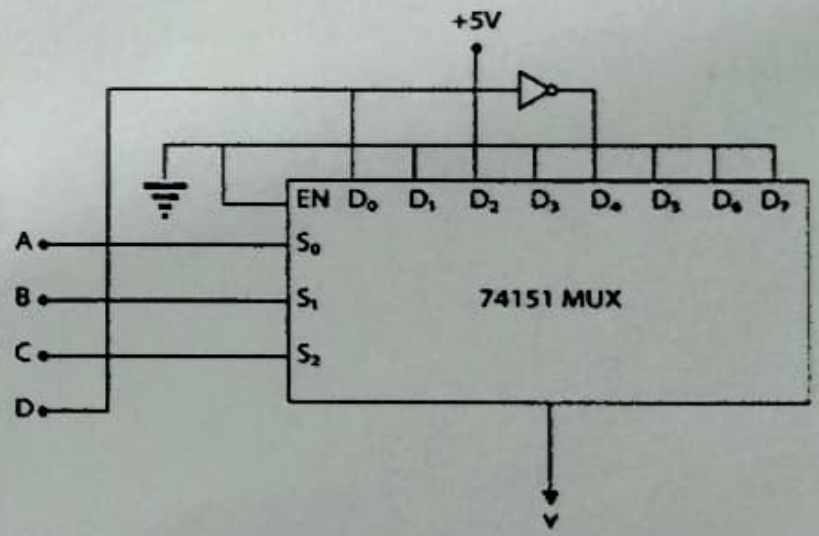
27. The circuit of figure shows how an 8-bit MUX can be used to generate a four variable logic function eventhough the MUX has only 3 SELECT inputs. Three of the logic variable A,B and C are connected to the SELECT inputs. **10**

The fourth variable D and its inverse $\overline{D}$ are connected to selected data inputs of the MUX as required by the desired logic function. The other MUX data inputs are used to a LOW or a HIGH as required by the function.

a) Setup a truth table showing the output Y for the 16 possible combinations of input variables.

b) Write the sum-of-products expression for Y and simplify it to verify that

$$Y = \overline{C}B\overline{A} + D\overline{C}\,\overline{B}\,\overline{A} + \overline{D}C\,\overline{B}\,\overline{A}$$



28. Simplify the Boolean function F using the don't care conditions d, in (1) sum of products and (2) product of sums: **10**

$$F = A'B'D' + A'CD + A'BC$$
$$d = A'BC'D + ACD + AB'D'$$

29. Given the following truth table: **10**

x	y	z	F_1	F_2	
0	0	0	0	0	0
0	0	1	1	0	1
0	1	0	1	0	2
0	1	1	0	1	3
1	0	0	1	0	4
1	0	1	0	1	5
1	1	0	0	1	6
1	1	1	1	1	7

(a) Express F_1 and F_2 in product of maxterms.
(b) Obtain the simplified functions in sum of products.
(c) Obtain the simplified functions in product of sums.

30. Explain the BCD-to-decimal decoder. **10**

31. Consider a JK' flip-flop, i.e., a JK flip-flop with an inverter between the external input K' and internal input K. **10**
(a) Obtain the flip-flop characteristic table.
(b) Obtain the characteristic equation.
(c) Show that tying the two external inputs together forms a D flip-flop.

32. Design a synchronous counter up-down with JK flip-flop which counts the odd number from 1 to 15 in binary. **10**

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